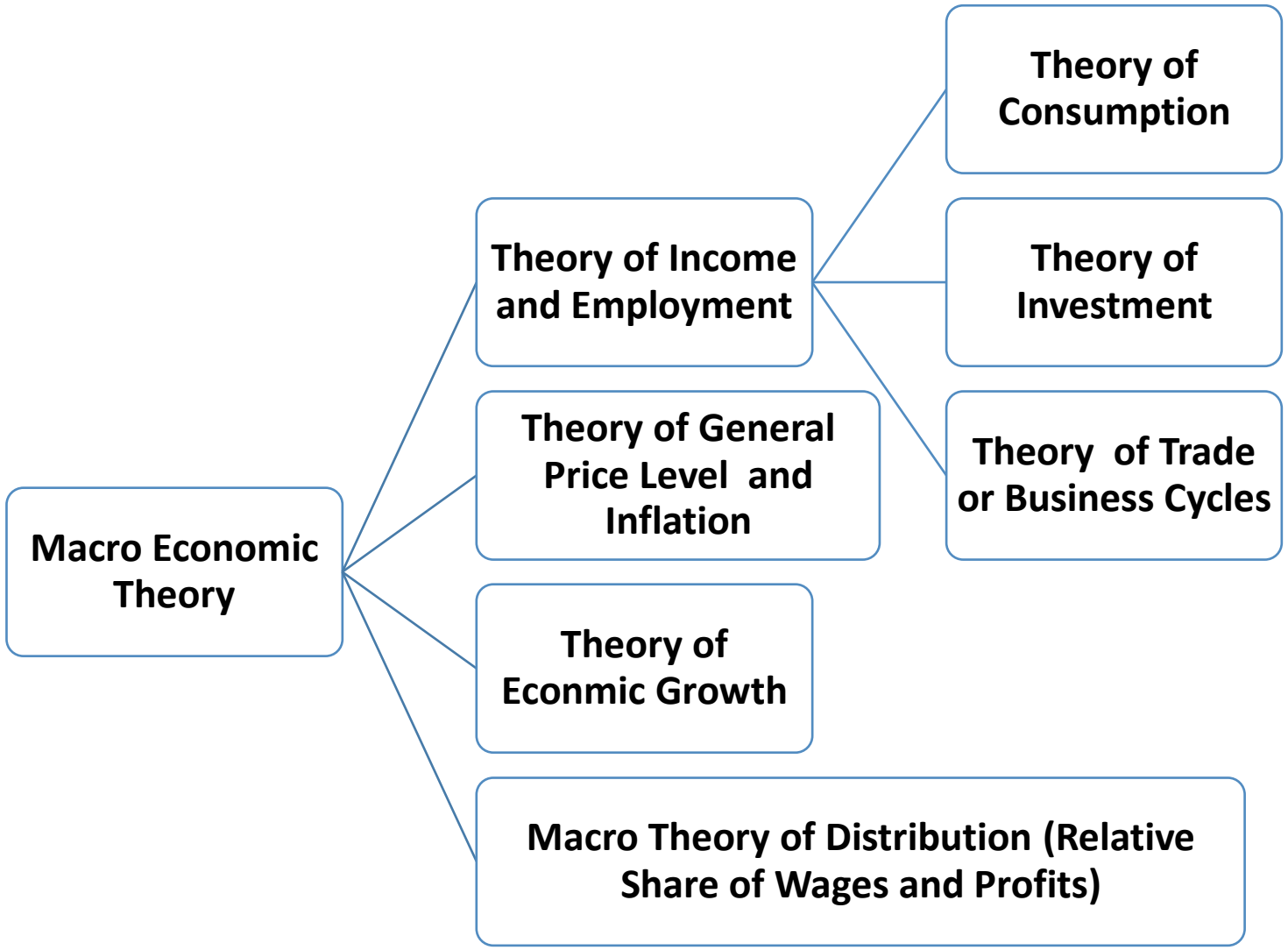
Macroeconmic Environment

Samir K Mahajan

MACROECONOMICS

***Macroeconomics* looks at the economy as an organic whole.** **Macro economics studies economic aggregates such as: total output, total demand, aggregate income, total savings, total investment, total employment, rise and fall in general price level, interest rates.**



ECONOMY AND BASIC ECONOMIC ACTIVITIES

Economic agents:

- **Individuals or Households (has dual roles: consumers and factor owners)**
- **Producers or Firms or Enterprises**
- **Government**
- **Foreigners**

RESOURCES / MEANS OF PRODUCTION/ FACTORS OF PRODUCTION / INPUTS OF PRODUCTION

Resources are the inputs into the production of goods and services which includes the followings:

- **Human Resources: *Labour and Entrepreneurship*. It All forms of human input, both physical and mental, into current production. The labour force is limited both in number and in skills.**

- **Natural Resources: *Land, mineral resources and Raw Materials*. These are inputs into production that are provided by nature: e.g. unimproved land and mineral deposits in the ground. The world's land area is limited, as are its raw materials.**

- **Manufactured Resources: *Capital*. Capital consists of all those inputs in production that have themselves been produced: e.g. factories, machines, transport equipments and tools. The world has a limited stock/supply of capital: The productivity of capital is limited by the state of technology.**

Thus, Resources are scarce in supply.

ECONOMY AND BASIC ECONOMIC ACTIVITIES

An economy is the system of earning livelihood (Brown). An economy is just a group of people dealing with one another as they go about their lives. Another more common definition: Economy is a system of four basic economic activities such as: production, consumption, investment and distribution of goods and services.

Basic economic activities production, distribution, consumption and investment are the vital process or necessary economic activities without which an economy cannot exist.

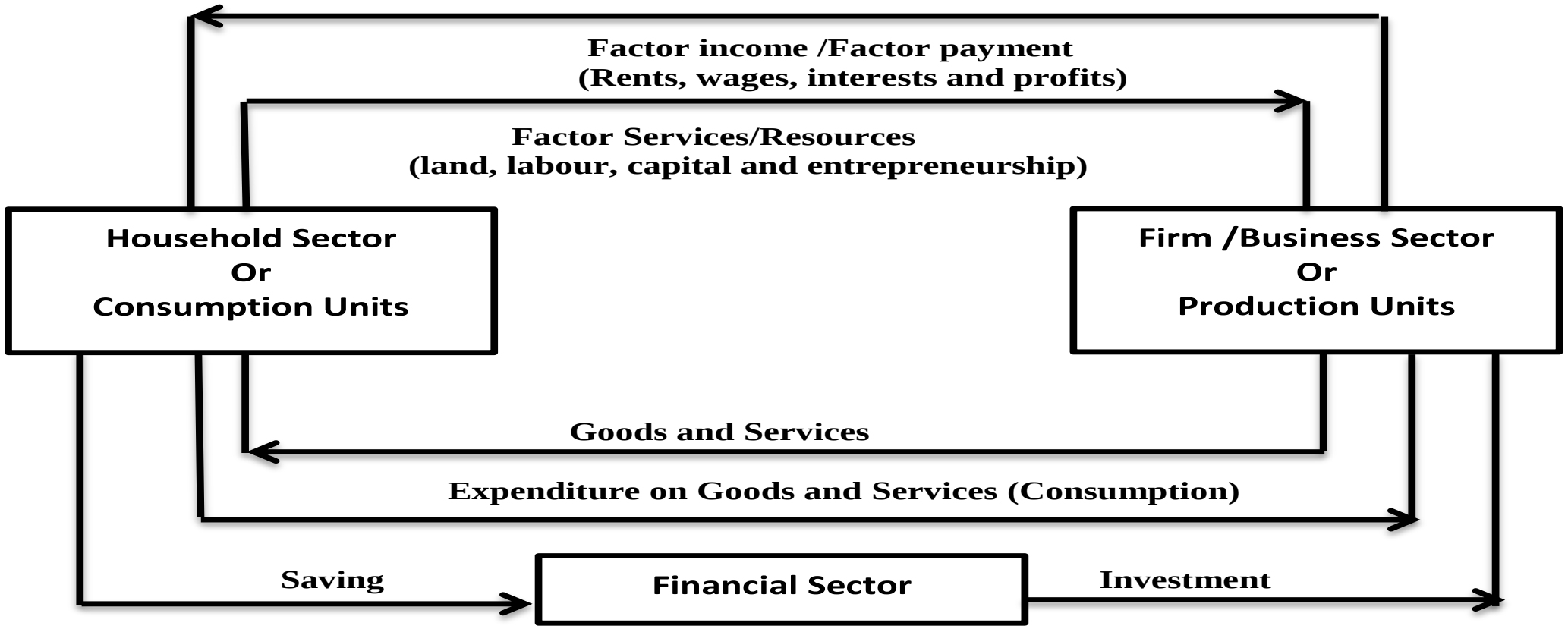
- **Production is transformation of inputs into output/finished products. It is creation or addition of utilities. We can not produce matter. Matters are free gift of nature. We make them more useful by transforming them into finished goods.**

- **Consumption is use of goods and services. It is destruction or decrease in utility in a particular commodity.**

- **Investment also called captain formation is the production of new capital goods. It is addition to existing capital stock of a society or economy. Investment is Excess of production over consumption which is used for further productive activities during an accounting period.**

- **Distribution is the sharing of produced goods and services or aggregate income among the various owners of factors of production in the form of rents, wages, interests and profits.**

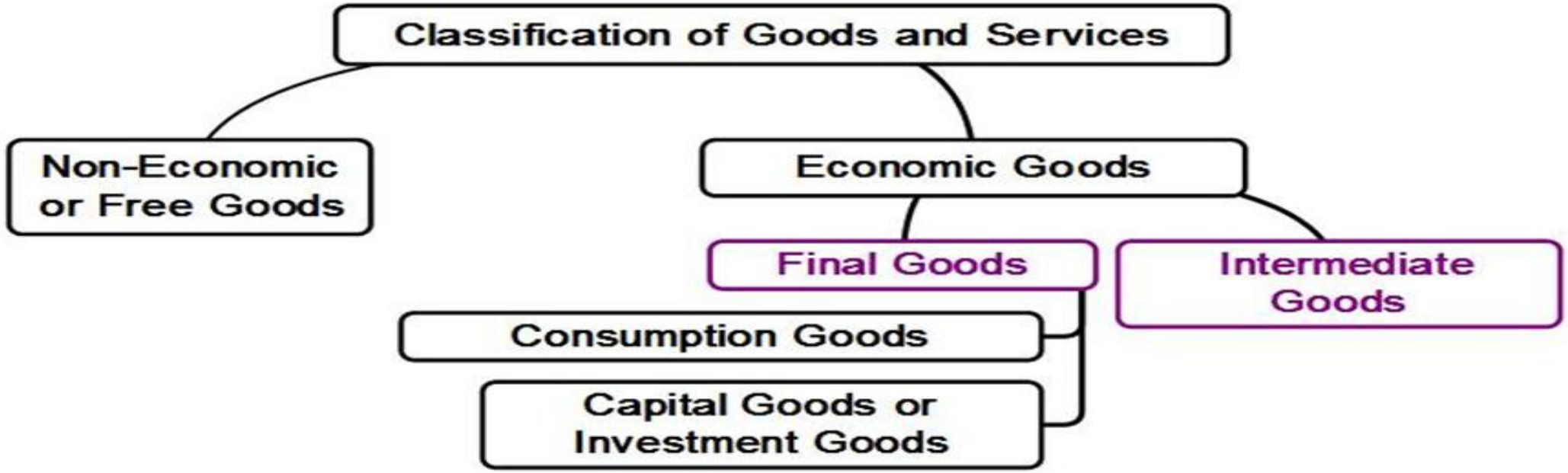
CIRCULAR FLOW (TWO SECTOR MODEL)



Aggregate Output = Aggregate Factor Income = Consumption +Saving = Consumption + Investment = Aggregate Expenditure

Samir K Mahajan

Classification of Goods and Services



Some Basic Concepts

- **Consumption Goods:** Consumption Goods are goods and services used directly by individuals/households for satisfaction of wants.

- **Capital Goods/investment goods:** Investment/capital goods are goods used in the manufacture of other goods. These are goods used for future/further productive activities such as roads, buildings, roads and bridges, transport equipments

- **Intermediate goods:** Intermediate goods are goods that have to pass through further production process or meant for resale during an accounting year. These are goods meant neither for consumption nor for investment. e.g. raw materials, fuels, electricity etc. Intermediate goods are not considered in the calculation of national income.

- **Final goods:** Final goods consists of consumption goods and investment goods. Final goods are considered in the calculation of national income.

- **Producer goods:** Producer goods consists of capital goods and intermediate goods.

- **Concepts of Capital formation or investment:**

- **Net Domestic Fixed investment or capital formation (I_{nf}) is the value of newly produced fixed capital goods i.e. assets within the country machineries, transport equipments, buildings, factories, plants, infrastructure such as: infrastructure such roads, buildings, roads and bridges.**

- **Replacement investment (D) is investment made to compensate depreciation. Deprecation is decrease in the value of fixed capital stock i.e. wear and tears in existing capital goods.**

- **Change in stock is the change in stock of inventories of raw materials, finished and semi-finished goods lying with the producers at the end of accounting year**

❑ Concepts of Capital Formation or Investment:

Equation of Other related concepts:

Net domestic capital formation = Net Domestic Fixed investment + Change in stock

Gross Fixed Capital formation (I_{gf}) = Net Domestic Fixed investment + Replacement investment or Depreciation

Thus,

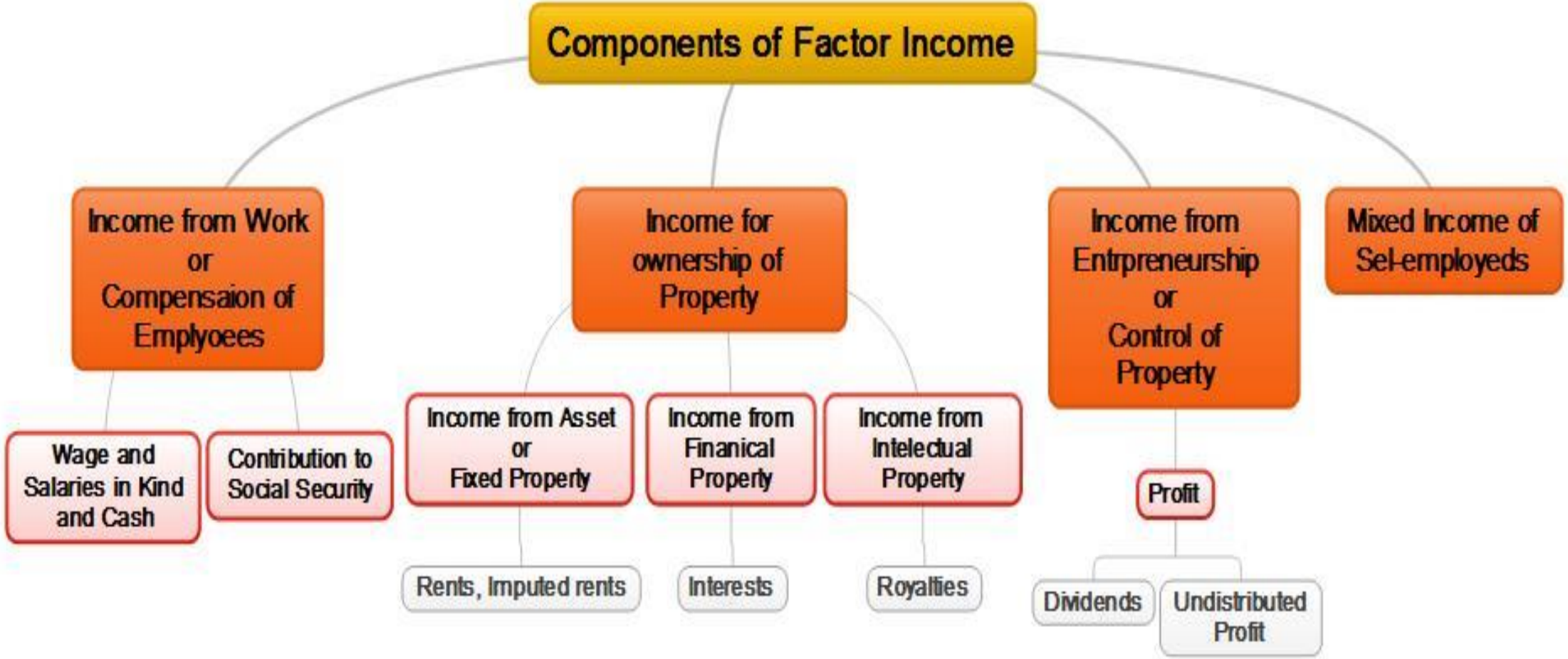
Gross domestic Capital formation (I_g) = Net Domestic Fixed Capital formation (I_{nf}) + Depreciation + Change in stock

= Gross Domestic Fixed Capital formation (I_{gf}) + Change in stock

= Net Domestic Capital formation (I_n) + Depreciation

- **Export:** Export is s ale of goods and services to foreign countries.
- Import: Import is purchase of goods and services from foreign countries.
- **Net Export:** = Export – Import
- *Saving : Saving* is income not spent, or deferred consumption. Saving = Income –consumption
- **Net factor income from Abroad (NFIA) =** factor income received from abroad – factor income paid to abroad.
- **Direct Taxes:** Direct Taxes are taxes imposed on income and capital.
- **Indirect taxes:** Indirect taxes are Taxes imposed on goods and services.
- **Subsidies:** Subsidies are economic grants provided to certain cover up the losses suffered when cost of production is more than the market prices.
- **Net Indirect Tax** = Indirect Tax – Subsidies
- **Transfer payments or non-factor income:** Transfer payments are payments received without contributing anything to production process e.g. unemployment benefits, old-age benefits, taxes, windfall profits from speculation activities say lottery , pocket money, gifts etc.

Factor Income: Factor incomes are the incomes generated and earned by the owners of factors of production. These includes wages, rents, interest, profits etc.



Producing Sector of the Economy : In order to determine the output of the economy , the economy is classified into three broad industrial sectors such as primary sector , secondary sector and tertiary sector.

- **Primary sector includes**
  - Agriculture
  - Mining and quarrying
  - Fishing and Animal Husbandry

- **Secondary Sector which includes**
  - Construction
  - Manufacturing

- **Tertiary sectors which includes**
  - Transportation, communication, storage and other public utilities
  - Trade
  - Hotels and restaurants
  - Banking , insurance and finance
  - Professions and business services including accounting; software development; data processing services; business and management consultancy; architectural, engineering and other technical consultancy; advertisement and other business services
  - real estate and Ownership of dwellings
  - Public administration and defence
  - Other services including education, medical and health, religious and other community services, legal services, recreation and entertainment services

CONCEPTS OF NATIONAL INCOME

❑ Gross Domestic Product (GDP) at market price:

Gross Domestic Product (GDP) at market price is the total money value of all final goods and services currently produced within the domestic territory of a country annually. Algebraic expression of GDP under output method is

GDP at Market Price = $\sum P * Q = \sum Value\ Added$

Where, P= Market price of each final good or service
Q=Quantity of each final good or service produced

GDP has the following components

- Value of final consumer goods and services produced during an accounting year (C)
- Gross domestic private capital formation or investment during an accounting year (I_g)
- Value of purchases of goods and services by government during the accounting year (G)
- Net Export = Export (X) – Import (M) during an accounting year

Thus, GDP at Market price = C + I_g + G + (X-M)

❑ Net Domestic Product (NDP) at market price:

Net Domestic Product at market price is the money value of all final goods and services produced within the country after providing for depreciation.

Algebraically, NDP at market = GDP at Market price – Depreciation

$$= C + I_g + G + (X-M) - D$$

$$= C + (I_g - D) + G + (X-M)$$

$$= C + I_n + G + (X-M)$$

Here, D is deprecation

I_n is net domestic capital formation or investment

❑ Gross National Product (GNP) at market price:

Gross National Product at market price is total money value of all final goods and services produced annually in a country plus net factor income from abroad.

Thus, GNP at Market price = GDP at Market price + NFIA

$$= C + I_g + G + (X-M) + NFIA$$

Where, NFIA is net foreign income from abroad

❑ Net National Product (NNP) at market price:

Net National Product at market price is the money value of all final goods and services of a country after providing for depreciation.

Thus, NNP at Market price= GNP at market price – Deprecation

= NDP at market price + NFIA

$$= C + Ig + G + (X\text{-}M) + NFIA - D$$
$$= C + (Ig - D) + G + (X\text{-}M) + NFIA$$
$$= C + I_n + G + (X\text{-}M) + NFIA$$

Here, D is deprecation

I_n is net domestic capital formation or investment

❑ Domestic Factor Income or Net Domestic Product at Factor Cost

Domestic Factor Income or Net Domestic product at factor cost is the sum total factor income generated and earned by suppliers/owners of factors of production within the domestic territory of country during a year.

Domestic Factor Income or NDP at factor cost = Wages and salaries in kind and cash + contribution to social security
+ Rents including imputed rents + royalties + Interests
+ undistributed Profits + dividends
+ Mixed incomes of self-employed

❑ National Income (NI) or NNP at Factor Cost :

National Income (NI) or NNP at Factor Cost is the sum total factor income generated and earned by suppliers/owners of factors of production in a country during a year.

National income or NNP at factor cost = Domestic Factor Income + Net factor income from abroad

= Wages and salaries in kind and cash + contribution to social security
+ Rents including imputed rents + royalties + Interests
+ undistributed Profits + dividends
+ Mixed incomes of self-employed
+ Net factor income from abroad

- **Gorss Domestic Factor Income or F Gross Domestic Product at Factor Cost**

= Domestic factor Income or NDP at factor cost + depreciation

Domestic Factor Income or NDP at factor cost = Wages and salaries in kind and cash + contribution to social security
+ Rents including imputed rents + royalties + Interests
+ undistributed Profits + dividends
+ Mixed incomes of self-employed
+ depreciation

- **Gross National Income (NI) or GNP at Factor Cost = National income or NNP at factor cost + Depreciation**

= Domestic factor Income or NDP at factor cost
+ net factor income from abroad + depreciation

❑ Relationship between NDP at market price and NDP at factor Cost

Had there been no government intervention in the economy, both NDP at market price NDP at factor Cost would have been equal to each other. As the government interferes with the economy through imposition of indirect taxes on goods and services, and provision of subsidies to productive enterprises, these cause market prices of output to be different from the factor income resulting from it. Hence, there arises difference between NDP at market price and NI (NDP at factor cost).

In accounting sense,

NDP at factor cost - subsidies = NDP at market price – indirect taxes

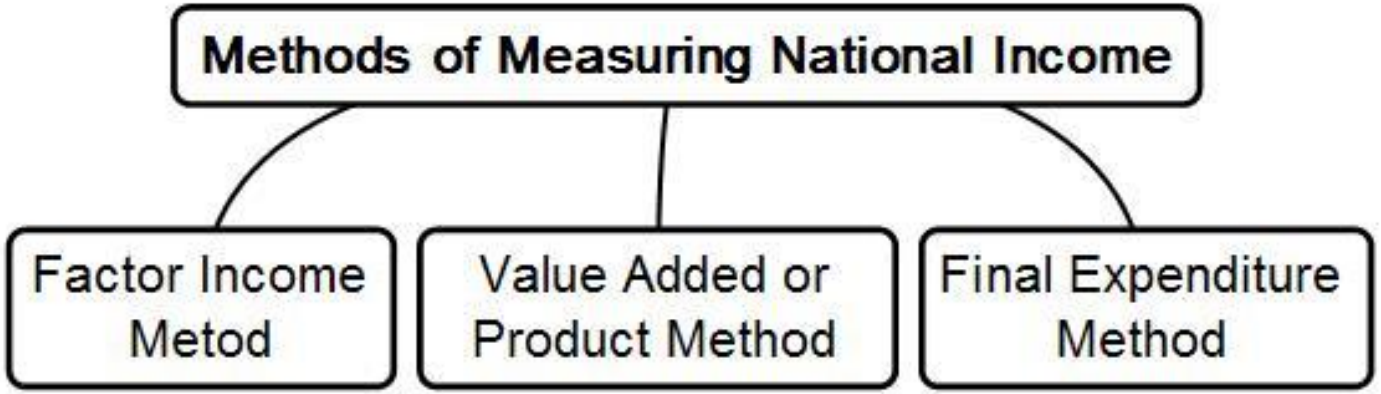
NDP at factor cost = NDP at market price – indirect taxes + subsidies

= NDP at market price – net indirect tax

Similarly,

- **GDP at factor cost = GDP at market price - indirect tax + subsidies**
- **NNP at factor cost = NNP at market price - indirect tax + subsidies**
- **GNP at factor cost = GNP at market price – indirect taxes + subsidies**

METHOD OF MEASUREMENT OF NATIONAL INCOME



❑ Value added Method or Product Method of Measuring National Income

Value added method adds up valued added by all productive sectors of the sectors of the economy annually. It measures final money value of all goods and services produced in a country during a year.

The mains steps involved in value method are as follows:

Step I: Identification and classification of production units located within the economic territories into three distinct industrial sectors on activity basis such as: primary, secondary and tertiary sectors. Final value of goods and services produced by these sectors are measured.

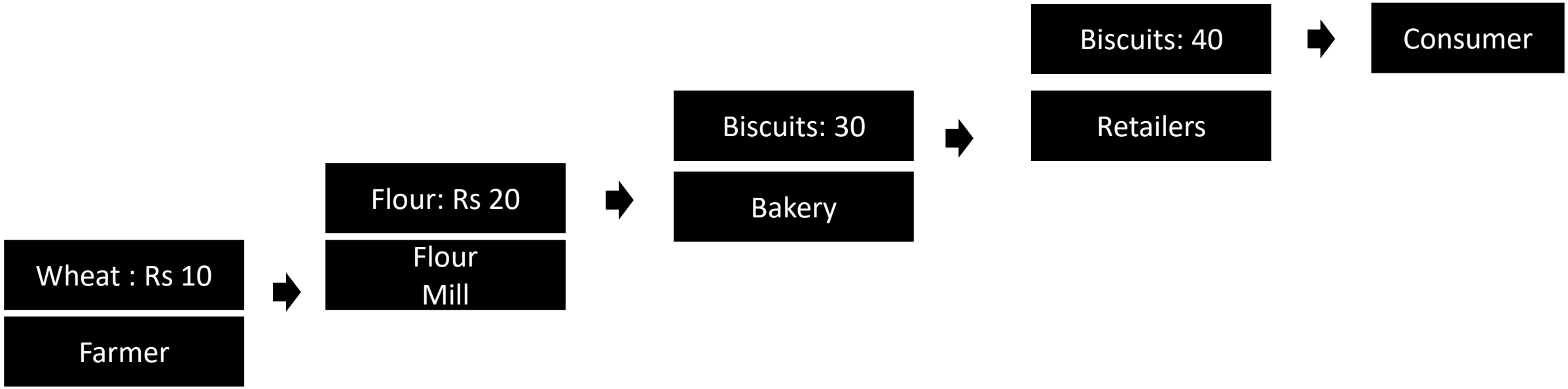
Step II: Estimation of Value Added by each production units in the industrial sector.

For this the following information are collected

- **Value of output**
- **Intermediate consumption**

Now, valued added = Value of Output – Intermediate consumption

Where , Value of output = Sales + Changes in stock+ Export
Intermediate consumption = Purchase of intermeddle goods fro domestic firms + imports of raw materials


Wheat : Rs 10
Farmer
Flour: Rs 20
Flour Mill
Biscuits: 30
Bakery
Biscuits: 40
Retailers
Consumer

Value added Method or Product Method contd.

Step III: Estimation of GNP at market price. Value added by all the production units in different industrial sectors is added to ascertain GDP at market price. Adding NFIA to GPP at Market price, we arrive at GNP at market price.

GDP at market price = $\sum Value\ added$

Adding NFIA with GDP at market price, we have GNP at market price

i.e. GDP at market price + NFIA = GNP at market price

Note: Net value added = Gross Value added – Depreciation

NDP at market price = $\sum Net\ Value\ added$

❑ Income Method of Measuring National Income

Income Method sums up all the factor incomes earned by suppliers /owners of factors of production annually. Factor of productions are engaged in production process and remunerated for their contribution.

The mains steps involved in production method are as follows:

Step I: Identification and classification of production units located within the economic territories into three distinct industrial sectors on activity basis such as: primary, secondary and tertiary sectors.

Step II: Classification of factor income earned by each factor separately in different sector of the economy. Factor incomes includes

- **income from work which includes wages and salaries in kind, contribution to social security**
- **income from ownership which includes rents, royalties and interests**
- **Income from control of property i.e. profits which includes dividends and undistributed profits**
- **mixed incomes of self employed**
- **net factor income from abroad (NFIA)**

Income Method contd.

Step III: Estimation of National Income. Factor incomes paid out by each industrial sector are ascertained. Summation of factors income paid out by all the industrial sectors NDP at factor cost. By adding NFIA to NDP at factor cost, we arrive at national income.

i.e. NDP at factor cost or Domestic Factor income= wages and salaries in kind and cash + contributions to social security
+ rents including imputed rents + royalties + interests + dividends
+ undistributed profits + mixed incomes

Adding NFIA with NDP at factor cost, we arrive at NNP at factor cost or national income.

i.e. NDP at factor cost + NFIA = NNP at factor cost or National Income

Final Expenditure Method of Measuring National Income

Final Expenditure Method sums up expenditure incurred by different economic units on final output (final goods and service) of the economy annually.

The mains steps involved in final expenditure are as follows:

Step I: Identification and classicisation of economic units into household sector , producer sector, government sector and rest of the world sector.

Step II: Classification of final expenditure. Final expenditure are classified as

- **Final consumption expenditure by household sector**
- **Gross domestic capital formation by producer sector**
- **Government final expenditure**
- **Net exports =Export –Import**

Step III: Estimation of Gross Domestic Expenditure (GDE) and Gross National Expenditure (GNE). Final expenditures are summed up to arrive at GDE . GDE is equivalent to GDP at market price.

Thus, GDE= GDP at market price = Private final consumption expenditure
+ Gross domestic private capital formation
+ Government final expenditure
+ Net Exports

Adding net factor income from abroad (NFIA) to GDE, we arrive at GNE or GNP at market price.

That i.e. , GNE = GNP at market price = GDE + NFIA

OTHER RELATED CONCEPTS

- **Aggregate Supply**

Aggregate supply, also known as total output, is the total supply of goods and services produced within an economy at a given in a given period. It is a macro-economic term that describes the relationship between price level and the quantity of output that firms are willing to provide.

- **Aggregate Demand**

Aggregate demand is an economic measurement of demand for the sum of all final goods and services produced in an economy in a given period. Aggregate demand is measured by market values. It's a macroeconomic term that describes the relationship between everything bought within a country and prices.

Aggregate Demand = Private final consumption expenditure
+ Private Domestic Investment Expenditure
+ Government expenditure on public goods and services
+ Net Exports

OTHER RELATED CONCEPTS

❑ FORMS OF MONEY IN MODERN TIMES

In modern monetary transactions, In modern era, the total stock of money or money supply includes the following:

- **Metallic money or currency coins : Metallic money refers to the coins made out of metal like gold, bronze, silver, copper, nickel. Coins are of two types such as: Standard or Full-bodied Coins and Token coins.**

  - **Standard or Full-bodied Coins are those coins whose face value is equal to its intrinsic (metallic) value.**

  - **Token coins have intrinsic (metallic) value less than its face value. They generally are of lower denominations are made of cheap metals like nickel and copper. Token coins are used for exchange of small value.**

- **Paper money or paper currency : Paper money consists of currency notes issued by the State Treasury or the Central bank of the country.**

- **Credit or bank money: Bank demand deposits withdraw-able by issuing cheque has started functioning as money, and cheque are conventionally accepted as a means of payment by the business community in general.**

OTHER RELATED CONCEPTS

Forms of money in modern times continued

In modern era, currency money and bank money together constitute the total stock of money or money supply.

- **Currency money (both currency coins and currency notes) is legal tender money or fiat money and has general acceptability.**

- **Credit money or bank demand deposits are conventional money and lacks general acceptability.**